



Basics of Quantum Information Theory and Quantum Tangent Kernel

DA6300: Quantum Computing and Machine Learning

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Example of Areas Where Quantum Computers Excel

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- **RSA Cryptosystem:** RSA is a widely-used public-key cryptosystem that relies on the difficulty of factoring large numbers as the foundation of its security. Classical computers would need an impractical amount of time to break RSA encryption by factoring large prime numbers.

For more details on Cryptography and RSA:

https://mdkarimullahaque.github.io/talks/Glimpses_of_Cryptography.pdf



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Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Shor's Algorithm:** Shor's Algorithm is a quantum algorithm that can efficiently factor large numbers in polynomial time, making it much faster than any classical algorithm. This breakthrough algorithm is the primary reason why quantum computers pose a threat to RSA encryption.

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Example of Areas Where Quantum Computers Excel

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Classical vs Quantum:** For classical computers, factoring a large number, such as those used in RSA keys (often hundreds of digits), is computationally infeasible and would take billions of years to break with current technology. In contrast, a quantum computer using Shor's Algorithm could factor these numbers in a matter of seconds or minutes.

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Cost of Building a Quantum Computer

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Hardware:** Quantum computers require special environments to operate, such as ultra-low temperatures (near absolute zero) for certain types of qubits like superconducting qubits.

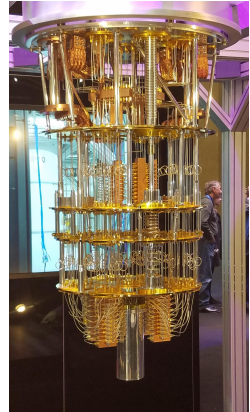


Figure: IBM Q quantum computer



Cost of Building a Quantum Computer

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Development:** Quantum computing is still in the experimental phase, meaning significant resources are required to build and maintain these systems.

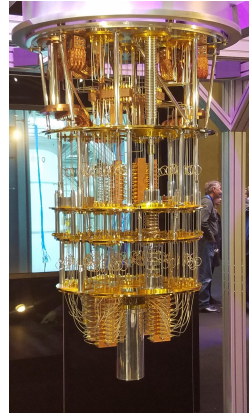


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Cost of Building a Quantum Computer

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Development:** Quantum computing is still in the experimental phase, meaning significant resources are required to build and maintain these systems.
- Overall, the cost of building a quantum computer is in the range of **tens of millions of dollars**, and the technology is still in a nascent stage, meaning the cost could remain high for the next several years until it matures.

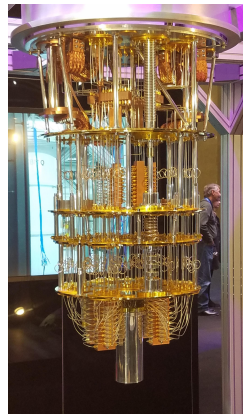


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Agenda

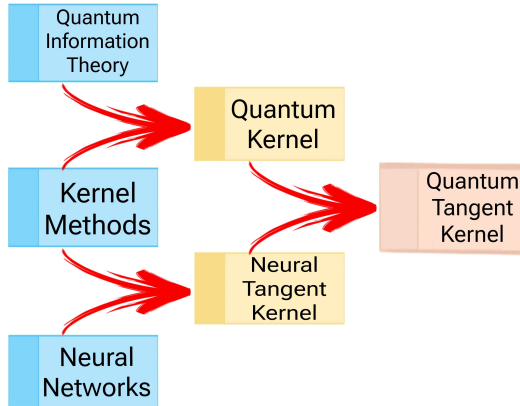
Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References



Agenda

Quantum Information Theory

Introduction

Superposition

Bloch Sphere

Entangled States

Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method

Neural Tangent Kernel

Quantum Tangent Kernel

Numerical Experiment

References

■ Qubits



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

■ Qubits

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Basic Terminology

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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Basic Terminology

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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Basic Terminology

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- **Ket:** $|*\rangle$, generally it's a column vector.



Basic Terminology

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

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Basic Terminology

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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Basic Terminology

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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Basic Terminology

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

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■ $|*\rangle \otimes |*\rangle = |*\rangle|*\rangle = |**\rangle$



Basic Terminology

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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■ $|*\rangle \otimes |*\rangle = |*\rangle|*\rangle = |**\rangle$

■ **Vector Notations:** $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, standard Qubits. $\langle 0| = (1 \ 0)$ and $\langle 1| = (0 \ 1)$



Agenda

Quantum Information Theory

Introduction

Superposition

Bloch Sphere

Entangled States

Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method

Neural Tangent Kernel

Quantum Tangent Kernel

Numerical Experiment

References

$$\blacksquare \langle a||b\rangle = \langle a|b\rangle = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases}$$

■ $|a\rangle\langle b|$ has a 1 in the (a,b)-entry and 0 for all other entries.



Agenda

Quantum Information Theory

Introduction

Superposition

Bloch Sphere

Entangled States

Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method

Neural Tangent Kernel

Quantum Tangent Kernel

Numerical Experiment

References

$$\blacksquare \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \text{ and } \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Let A be an $m \times n$ matrix and B be a $p \times q$ matrix then



Tensor Product

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Let A be an $m \times n$ matrix and B be a $p \times q$ matrix then

- $$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}$$



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Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- is an $(mp) \times (nq)$ matrix.



Tensor Product

Agenda

Quantum Information Theory

Introduction

Superposition

Bloch Sphere

Entangled States

Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method

Neural Tangent Kernel

Quantum Tangent Kernel

Numerical Experiment

References

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- is an $(mp) \times (nq)$ matrix.

- Example:

- $\sigma_x \otimes \sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$



Superposition

Agenda

Quantum Information Theory

- Introduction
- Superposition**
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- It is convenient to assume the vector $|0\rangle$ corresponds to the classical value 0, while $|1\rangle$ to 1 in quantum computation. Moreover it is possible for a qubit to be in a superposition state:



Superposition

Agenda

Quantum Information Theory

- Introduction
- Superposition**
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\alpha, \beta \in \mathbb{C}$



Agenda

Quantum Information Theory

- Introduction
- Superposition**
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\alpha, \beta \in \mathbb{C}$
- The fundamental requirement of quantum mechanics is that if we make measurement on $|\psi\rangle$ to see whether it is in $|0\rangle$ or $|1\rangle$, the outcome will be 0 (1) with the probability $|\alpha|^2 (|\beta|^2)$,



Bloch Sphere

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- **Definition:** Quantum analog of classical bits; exists in superposition states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where

$\alpha, \beta \in \mathbb{C}$

$$= \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where $|\alpha|^2 + |\beta|^2 = 1$.

Let, $\alpha = r_1 e^{i\theta_1}$ and $\beta = r_2 e^{i\theta_2}$

Then, $r_1^2 + r_2^2 = 1$ and $\theta_1 = \theta_2 + \varphi$

Take, $r_1 = \cos\left(\frac{\theta}{2}\right)$, $r_2 = \sin\left(\frac{\theta}{2}\right)$

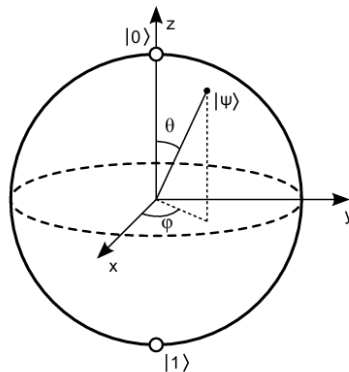


Figure: The Bloch sphere is a geometrical representation of a qubit. Qubits can take as value each point on the surface described by the two angles φ and θ . The pole points are $|0\rangle$ or $|1\rangle$.



Entangled States

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- A state $|\psi\rangle \in \mathcal{H}$ written as a tensor product of two vectors as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ (where $|\psi_a\rangle \in \mathcal{H}_a$) is called a **separable state** or a **tensor product state**.



Entangled States

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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- Let us consider a state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$ of two separated electrons. Suppose $|\psi\rangle$ may be decomposed as
$$|\psi\rangle = (c_1|0\rangle + c_2|1\rangle) \otimes (d_1|0\rangle + d_2|1\rangle) = c_1 d_1 |0\rangle \otimes |0\rangle + c_1 d_2 |0\rangle \otimes |1\rangle + c_2 d_1 |1\rangle \otimes |0\rangle + c_2 d_2 |1\rangle \otimes |1\rangle$$



Entangled States

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- However this decomposition is not possible since we must have $c_1d_2 = c_2d_1 = 0$, $c_1d_1 = c_2d_2 = \frac{1}{\sqrt{2}}$ simultaneously, and it is clear that the above equations have no common solution. Therefore the state $|\psi\rangle$ is not separable.



Entangled States

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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- Such non-separable states are called **entangled** in quantum theory.



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- we have introduced qubits to store information.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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Quantum Computation

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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- More complicated **quantum circuits** are composed of these simple gates.



Quantum Computation

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- A collection of quantum circuits for executing a complicated **quantum algorithm**



Quantum Computation

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- If they are simple, these operations are called gates, or more precisely **quantum gates**
- More complicated **quantum circuits** are composed of these simple gates.
- A collection of quantum circuits for executing a complicated **quantum algorithm**
- a quantum algorithm, is a part of a **quantum computation**.



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

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Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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- A register or a set of registers



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- A quantum computation is a collection of the following three elements:
- A register or a set of registers
- A unitary matrix U , which is tailored to execute a given quantum algorithm



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- A quantum computation is a collection of the following three elements:
- A register or a set of registers
- A unitary matrix U , which is tailored to execute a given quantum algorithm
- Measurements to extract information we need.



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Let us consider the gate I whose action on the basis vectors are defined by $I : |0\rangle \rightarrow |0\rangle, |1\rangle \rightarrow |1\rangle$



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

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- Let us consider the gate I whose action on the basis vectors are defined by $I : |0\rangle \rightarrow |0\rangle, |1\rangle \rightarrow |1\rangle$
- The matrix expression of this gate as

$$I = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



NOT, shift gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

■ Similarly we introduce

■ $X : |0\rangle \rightarrow |1\rangle, |1\rangle \rightarrow |0\rangle$

■ $Y : |0\rangle \rightarrow -|1\rangle, |1\rangle \rightarrow |0\rangle$

■ $Z : |0\rangle \rightarrow |0\rangle, |1\rangle \rightarrow -|1\rangle$



NOT, shift gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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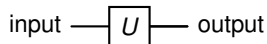
- $Z : |0\rangle \rightarrow |0\rangle, |1\rangle \rightarrow -|1\rangle$

- whose matrix representations are

$$X = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$$

$$Y = |0\rangle\langle 1| - |1\rangle\langle 0| = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z$$



CNOT (controlled-NOT) gate

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- It is a two-qubit gate



CNOT (controlled-NOT) gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- It is a two-qubit gate
- The gate flips the second qubit (the target qubit) when the first qubit (the control qubit) is $|1\rangle$, while leaving the second bit unchanged when the first qubit state is $|0\rangle$.



CNOT (controlled-NOT) gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- It is a two-qubit gate
- The gate flips the second qubit (the target qubit) when the first qubit (the control qubit) is $|1\rangle$, while leaving the second bit unchanged when the first qubit state is $|0\rangle$.
- Let $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ be a basis for the two-qubit system.



CNOT (controlled-NOT) gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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$$\text{■ } |00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$



CNOT (controlled-NOT) gate

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- It is a two-qubit gate
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- $|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $|11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

- $U_{CNOT} : |00\rangle \rightarrow |00\rangle, |01\rangle \rightarrow |01\rangle, |10\rangle \rightarrow |11\rangle$ and $|11\rangle \rightarrow |10\rangle$ In the following, we use the standard basis vectors with components



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- whose matrix representation is

$$\begin{aligned} U_{CNOT} &= |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \\ &= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{aligned}$$



Agenda

Quantum Information Theory

Introduction
 Superposition
 Bloch Sphere
 Entangled States
 Quantum Computation

Quantum Tangent Kernel

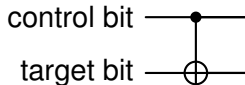
Kernel Methods
 QML as a Kernel Method
 Neural Tangent Kernel
 Quantum Tangent Kernel
 Numerical Experiment

References

- whose matrix representation is

$$\begin{aligned}
 U_{CNOT} &= |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \\
 &= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}
 \end{aligned}$$

- The action of CNOT on the input state $|i\rangle|j\rangle$ is written as $|i\rangle|i \oplus j\rangle$, where $i \oplus j$ is an addition mod 2, that is, $0 \oplus 0 = 0, 0 \oplus 1 = 1, 1 \oplus 0 = 1$ and $1 \oplus 1 = 0$.



Toffoli gate or, CCNOT(Controlled-Controlled-NOT) gate

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

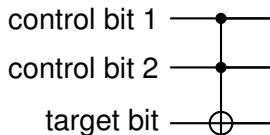
Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- The explicit form of the CCNOT gate is

$$U_{CCNOT} = (|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10|) \otimes I + |11\rangle\langle 11| \otimes X$$



Hadamard gate

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

■
$$U_H : |0\rangle \rightarrow |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), |1\rangle \rightarrow |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



Hadamard gate

Agenda

Quantum Information Theory

Introduction

Superposition

Bloch Sphere

Entangled States

Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method

Neural Tangent Kernel

Quantum Tangent Kernel

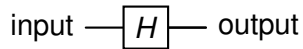
Numerical Experiment

References

■ $U_H : |0\rangle \rightarrow |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), |1\rangle \rightarrow |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

■ whose matrix representation is

$$U_H = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\langle 0| + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



Arithmetic Circuit

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

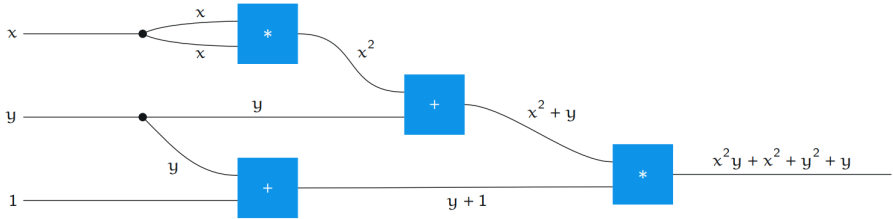


Figure: Wires store numbers and gates represent arithmetic operations, such as addition (+) and multiplication (*)



Example of Quantum Circuit

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

$$|0\rangle \xrightarrow{H} \xrightarrow{S} \xrightarrow{H} \xrightarrow{T} \frac{1+i}{2} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

■ where $H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$, $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$

We get $THSH = \begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix}$



Why Kernel?

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods

QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- Assume the feature map $\phi : \mathcal{X} \rightarrow \mathcal{F}$

- For example, $\phi \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} x_1 \\ x_2 \\ x_1 x_2 \end{pmatrix} \in \mathcal{F}$, where $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathcal{X}$

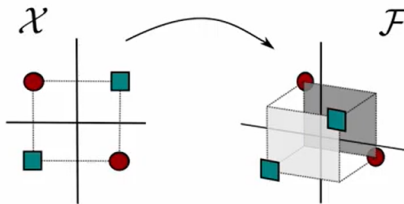


Figure: Separation may be easier in higher dimensions



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

Kernel Methods

- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Training dataset $\mathcal{D} = (x_i, y_i)_{i=1}^N$, where x_i is input data and y_i is corresponding teacher data.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- Training dataset $\mathcal{D} = (x_i, y_i)_{i=1}^N$, where x_i is input data and y_i is corresponding teacher data.
- Kernel method employs a non-linear map ϕ from the original space to the higher dimensional feature space: $\phi : \mathcal{X} \rightarrow \mathcal{F}$,
i.e., $x_i \rightarrow \phi(x_i)$,
where $\mathcal{X}$ is an original space of the data and $\mathcal{F}$ is a higher dimensional feature space.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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i.e., $x_i \rightarrow \phi(x_i)$,
where $\mathcal{X}$ is an original space of the data and $\mathcal{F}$ is a higher dimensional feature space.
- For two data x_i and x_j , we define $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$, where $\langle \cdot, \cdot \rangle$ denotes the inner product on $\mathcal{F}$.



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- First, data are encoded into quantum states by $U_\phi(x)$
i.e., $y(x, \theta) = \langle 0^n | U^\dagger(x, \theta) O U(x, \theta) | 0^n \rangle$
$$= \langle 0^n | U_\phi^\dagger(x) V^\dagger(\theta) O V(\theta) U_\phi(x) | 0^n \rangle \quad \{ \because U(x, \theta) = V(\theta) U_\phi(x) \}$$
$$= \text{Tr}(O(\theta) \rho(x)),$$
where weight vector $O(\theta) = V^\dagger(\theta) O V(\theta)$ and feature vector $\rho(x) = U_\phi(x) | 0^n \rangle \langle 0^n | U_\phi^\dagger(x)$



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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$$= \text{Tr}(O(\theta) \rho(x)),$$

where weight vector $O(\theta) = V^\dagger(\theta) O V(\theta)$ and feature vector $\rho(x) = U_\phi(x) | 0^n \rangle \langle 0^n | U_\phi^\dagger(x)$

- Now the conventional quantum kernel is defined as
 $K_q(x_i, x_j) = \text{Tr}(\rho(x_i) \rho(x_j)) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$, where $|\phi(x)\rangle = U_\phi(x) | 0^n \rangle$



Accuracy Compare

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

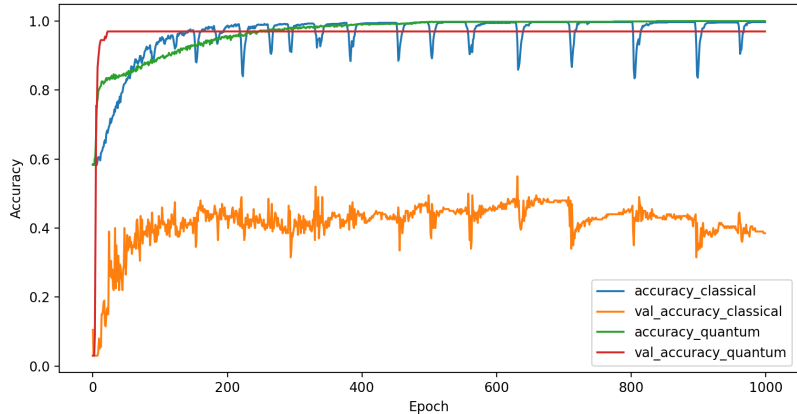


Figure: Comparison between Classical and Quantum Models

Neural Tangent Kernel (NTK)

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel**
- Quantum Tangent Kernel
- Numerical Experiment

References

- Suppose $f(x_i, \theta)$ is the output of a neural network where θ is parameters in the network, and x_i is the input data.



Neural Tangent Kernel (NTK)

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- Suppose $f(x_i, \theta)$ is the output of a neural network where θ is parameters in the network, and x_i is the input data.
- Now we approximated by the first-order expansion with respect to the parameters around the initial values:
$$f(x_i, \theta) \simeq f(x_i, \theta_0) + \nabla_{\theta} f(x_i, \theta_0)^T (\theta - \theta_0),$$
 where θ_0 is the initial values of parameters of a neural network.



Neural Tangent Kernel (NTK)

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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 where θ_0 is the initial values of parameters of a neural network.
- This approximation allows us to interpret the neural network as a linear model with a feature map $\phi(x) = \nabla_{\theta} f(x, \theta_0)$



Neural Tangent Kernel (NTK)

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- Now we approximated by the first-order expansion with respect to the parameters around the initial values:
$$f(x_i, \theta) \simeq f(x_i, \theta_0) + \nabla_{\theta} f(x_i, \theta_0)^T (\theta - \theta_0),$$
 where θ_0 is the initial values of parameters of a neural network.
- This approximation allows us to interpret the neural network as a linear model with a feature map $\phi(x) = \nabla_{\theta} f(x, \theta_0)$
- Using this feature map, we define the following neural tangent kernel (NTK):
$$K_{ntk}(x_i, x_j) = \nabla_{\theta} f(x_i, \theta_0)^T \nabla_{\theta} f(x_j, \theta_0)$$



Quantum Tangent Kernel (QTK)

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Now from K_q and K_{ntk} , we define the following quantum tangent kernel (QTK):

$$K_{qtk}(x_i, x_j) = \nabla_{\theta} y(x_i, \theta_0)^T \nabla_{\theta} y(x_j, \theta_0),$$

$$\text{where } y(x, \theta) = \langle 0^n | U^{\dagger}(x, \theta) O(\theta) U(x, \theta) | 0^n \rangle$$



Numerical Experiment

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- First one is a circuit where x is encoded only at the first layer as in the bellow Figure. $U_{\text{shallow}}(x, \theta) = V(\theta)U_{\phi}(x)$, where $V(\theta)$ is a parameterized unitary and $U_{\phi}(x)$ is a quantum feature map to encode data.

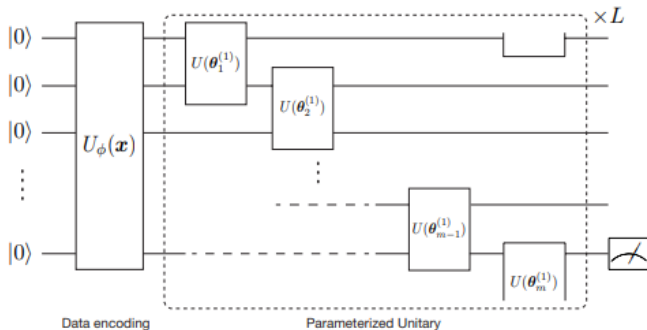


Figure: The m -qubit ansatz used for numerical simulations. $U_{\phi}(x)$ is the quantum feature map for encoding classical data. $U(\theta_i^{(1)}) \in SU(4)$ is the parameterized unitary of the i -th layer. At the output, we measure the Pauli Z expectation value of the final qubit.



Numerical Experiment

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Next, in order to increase the non-linearity, we consider a multi-layered circuit that alternates between data encoding and a parameterized unitary as shown in the bellow Figure. $U_{deep}(x, \theta) = \prod_{i=1}^L V(\theta_i) U_{\phi}(x)$,

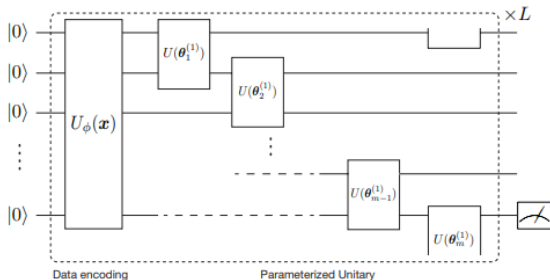


Figure: Quantum circuit that defines deep quantum tangent kernel. $U_{\phi}(x)$ is the feature map and $U(\theta_j^{(i)}) \in SU(4)$ is the parameterized unitary of the i -th layer. To increase the non-linearity of the kernel, the feature map $U_{\phi}(x)$ and the parameterized unitary are iteratively applied. At the output, we measure the Pauli Z expectation value of the final qubit.



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- The “ansatz-generated” dataset is generated in the following manner. Four dimensional random value data x_i are inputted into $U_{deep}(x, \theta)$ consisting of $n = 4$ qubits and $L = 10$ layers.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- Then, evaluate the expectation value $I(x_i, \theta) = \langle 0^n | U_{deep}^\dagger(x_i, \theta) Z_4 U_{deep}(x_i, \theta) | 0^n \rangle$ with a randomly chosen θ . We label each x_i as 1 and -1 if $I(x, \theta) \geq 0$ and $I(x, \theta) < 0$, respectively.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

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- Here, generate the 15,000 samples of the four dimensional input data x_i and its label y_i . They are splitted into 10,000 training and 5000 test data.



Behavior of training losses and training accuracies

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Behavior of training losses and training accuracies for different number of layers during training.

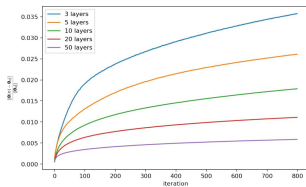


Figure: Relative norm change in the parameters of the quantum circuit from initial values during training by gradient descent. $\theta(n)$ is the parameter at the n -th iteration. θ_0 is initial value of the parameter.

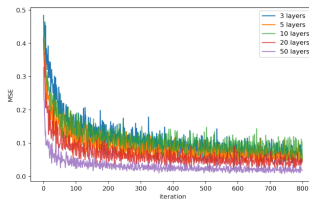


Figure: Mean Square Error

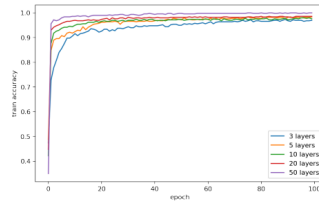


Figure: Train Accuracy



Classification accuracy for three types of kernels

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Classification accuracy for SVM with three types of kernels. Three SVMs classify the ansatz-generated dataset generated by a quantum circuit for deep QTK

Kernel	Accuracy
Quantum kernel	0.7842
Shallow quantum tangent kernel	0.7484
Deep quantum tangent kernel	0.812



The distribution of outputs

Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

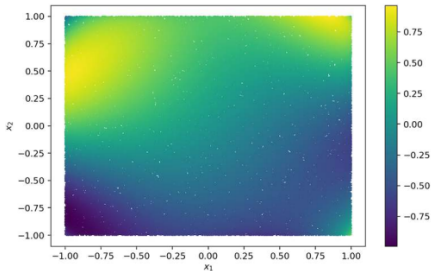


Figure: The distribution of outputs of the quantum circuit with $L = 10$ layers which can be interpreted as the conventional quantum kernel method.

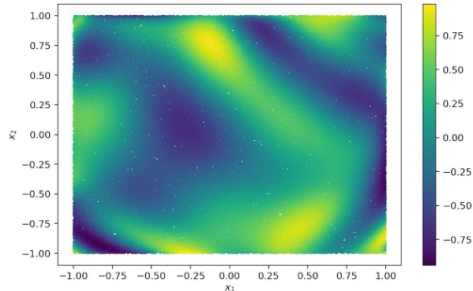


Figure: The distribution of outputs of the quantum circuit with $L = 10$ layers beyond the conventional quantum kernel



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Here quantum tangent kernel (QTK) and deep quantum tangent kernel which cannot be interpreted as conventional quantum kernel methods



Agenda

Quantum Information Theory

- Introduction
- Superposition
- Bloch Sphere
- Entangled States
- Quantum Computation

Quantum Tangent Kernel

- Kernel Methods
- QML as a Kernel Method
- Neural Tangent Kernel
- Quantum Tangent Kernel
- Numerical Experiment

References

- Here quantum tangent kernel (QTK) and deep quantum tangent kernel which cannot be interpreted as conventional quantum kernel methods
- QTK is defined by applying the formulation of NTK to parametrized quantum circuits.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- Here quantum tangent kernel (QTK) and deep quantum tangent kernel which cannot be interpreted as conventional quantum kernel methods
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- It imply that deep parameterized quantum circuits with repetitive data encoding unitary have a higher representation power and better performance for quantum machine learning than the conventional quantum kernel method.



Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

References

- Here quantum tangent kernel (QTK) and deep quantum tangent kernel which cannot be interpreted as conventional quantum kernel methods
- QTK is defined by applying the formulation of NTK to parametrized quantum circuits.
- It implies that deep parameterized quantum circuits with repetitive data encoding unitary have a higher representation power and better performance for quantum machine learning than the conventional quantum kernel method.
- Hence, better ansatz constructions and parameter optimization methods are crucially important.



References

Agenda

Quantum Information Theory

Introduction
Superposition
Bloch Sphere
Entangled States
Quantum Computation

Quantum Tangent Kernel

Kernel Methods
QML as a Kernel Method
Neural Tangent Kernel
Quantum Tangent Kernel
Numerical Experiment

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- Pictures were taken from Google Images.



Thank You

